

Software Project Engineering

A short, practical workflow for starting and designing a software project

Build Workflow

#	Step	What to do	Output
1	Problem	Define the problem you are solving.	Problem statement
2	Users	Identify who will use it.	Target users
3	Requirements	List the must-have features.	MVP requirements
4	Scope	Decide what is in/out of v1.	MVP scope
5	User Flow	Map how users move through the system.	User-flow diagram
6	System Design	Choose frontend, backend, database, APIs, auth, etc.	Architecture diagram
7	Data Design	Design entities, relationships, and database tables.	ERD / schema
8	API Design	Define endpoints, inputs, outputs, and permissions.	API specification
9	UI/UX	Design key screens and interactions.	Wireframes / UI
10	Tech Stack	Choose technologies based on the requirements.	Stack decision
11	Build Plan	Break the MVP into small tasks/issues.	Development roadmap
12	Implement	Build features incrementally.	Working features
13	Test	Test functionality, edge cases, security, and performance.	Tested MVP
14	Deploy	Set up the production environment and CI/CD.	Live application
15	Monitor & Improve	Track bugs, usage, feedback, and iterate.	v1.1 → v2

Problem → Requirements → Scope → Flow → Architecture → Data → API → UI → Build → Test → Deploy → Improve

Engineering principle: Design enough to make good decisions, then build. Avoid over-engineering before the MVP proves the idea.